

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ

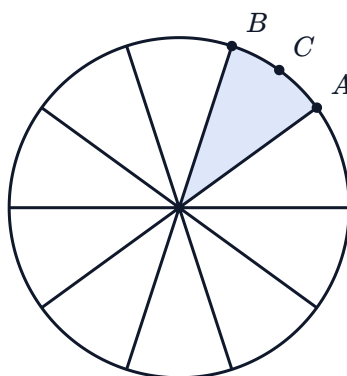
Chapter 11: Areas Related to Circles

Previous year board questions, topic-wise: 10 questions

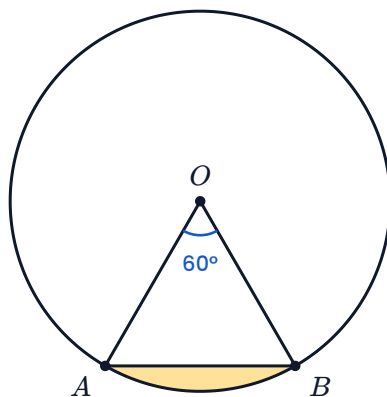
The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Sector and segment of a circle

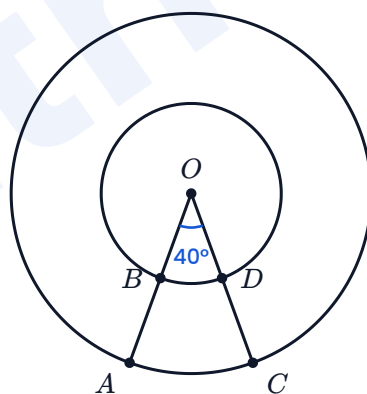
1. A sector of a circle of radius 21 cm has a central angle of 60° . Find the length of its arc. (a) 22 cm (b) 44 cm (c) 88 cm (d) 11 cm **CBSE 2026, 1 mark**
2. The hour hand of a clock is 7 cm long. Through what angle does it turn between 7:00 a.m. and 8:10 a.m.? (a) $\left(\frac{35}{4}\right)^\circ$ (b) $\left(\frac{35}{2}\right)^\circ$ (c) 35° (d) 70° **CBSE 2026, 1 mark**
3. A circular ornament of diameter 35 cm is made from silver wire. Wire is also used for 5 diameters, which cut the circle into 10 equal sectors. (i) Find the radius. (ii) Find the circumference. (iii) Find the total length of wire needed, OR the area of each sector. **CBSE 2026, 4 marks**
4. A sector with a central angle of 72° has an area of 40π square units. Find the radius of the circle. **CBSE 2025, 1 mark**
5. A wire 20 cm long is bent into an arc of a circle of radius $\frac{60}{\pi}$ cm. What angle does the arc subtend at the centre? **CBSE 2025, 1 mark**
6. A circular ornament of diameter 35 mm is made from silver wire. Wire also forms 5 diameters, which cut the circle into 10 equal sectors. A and B are the ends of one sector on the circle and C lies on the arc between them. Take $\pi = \frac{22}{7}$. (i) Find the central angle of each sector. (ii) Find the length of the arc ACB . (iii) Find the area of each sector, OR the total length of wire used. **CBSE 2025, 4 marks**



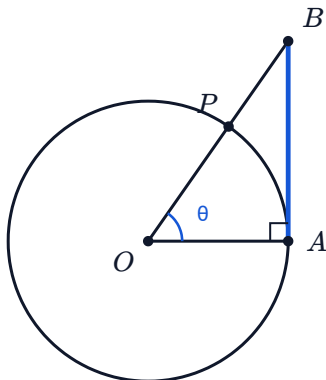
7. In a circle of radius 21 cm, an arc subtends 60° at the centre. Find (i) the length of the arc (ii) the area of the minor segment cut off by the chord of this arc. **CBSE 2024, 5 marks**
8. A chord of a circle of radius 14 cm subtends an angle of 60° at the centre. Find the area of the minor segment and the area of the major segment. Take $\pi = \frac{22}{7}$ and $\sqrt{3} = 1.732$. **CBSE 2023, 5 marks**



9. Two concentric circles with centre O have radii 7 cm and 14 cm. A sector of angle 40° is cut away from the ring between them. Find the area of the part of the ring that remains. Take $\pi = \frac{22}{7}$. **CBSE 2016, 3 marks**



10. OAP is a sector of a circle with centre O , radius r and angle θ (in degrees). AB is perpendicular to the radius OA and meets OP produced at B . Prove that the perimeter of the region bounded by AB , BP and the arc PA is $r \left[\tan \theta + \sec \theta + \frac{\pi \theta}{180} - 1 \right]$. **CBSE 2016, 4 marks**



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